Werfen Group | Instrumentation Laboratory

OPERATIONAL QUALIFICATION
IL GEM BLOOD GAS SYSTEMS

Prepared by:
Instrumentation Laboratory

Prepared for:
Facility name: Dushanbe International Hospital
Address: 734000, Kahhorov Str, 11, Dushanbe, Republic of Tajikistan

Equipment location: Laboratory

Date: 10.10.2026

Equipment installed

Model number: GemPremier 3500
Serial number: 26052266
Date of installation: 10.10.2026

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1.0 OQ Approval

The Operational Qualification has been prepared and approved by Instrumentation Laboratory Company.

Name / Position	Company	Date
Quality Assurance Engineering Manager	Instrumentation Laboratory	July 1, 2011
Director Customer Support	Instrumentation Laboratory	July 1, 2011

2.0 OQ Execution

The Operational Qualification has been conducted and completed by Instrumentation Laboratory Company or an authorized dealer.

Name / Position	Company	Date
Ilya Navopashin Engineer	Labtech Diagnostics	10.10.2026

3.0 OQ Documentation

A copy of the Operational Qualification documentation will be provided to;

Facility: Dushanbe International Hospital
Name: Partho Baruah
Position: Head of Laboratory

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4.0 OQ Scope

The scope of the Operational Qualification defines the requirements and criteria of acceptance which verifies the operation and functionality of the equipment. Equipment analytical performance verification will include successful cartridge warm-up, with all analytes in a "ready" state and successful analysis of the required levels of CVP. The steps outlined in this procedure shall be followed as written with results recorded. Any abnormalities along with corrective actions will be noted in the Operational Qualification documentation. Successful completion of this protocol ensures that this instrument is performing to Manufacturer's specifications.

5.0 Operational Qualification Requirements

5.1 Verify that Installation Qualification has been successfully completed.

Facility/Company	Title / Position	Name (print)	Signature
LabTech Diagnostics	Engineer	Ilya Novoposhin	

Notes: _____

5.2 Verify that the cartridge has completed warm up with all analytes in a "ready" state.;

Facility/Company	Title / Position	Name (print)	Signature
Labtech Diagnostics	Engineer	Ilya Novoposhin	

Notes: _____

5.3 Performance Checks

Verify that all required levels of CVP are within range.

CVP Level 1 Lot Number 1852.

Analyte	CVP Range	CVP Result	Corrective Actions Applied
pH	7,20 – 7,26	7,26	
PCO2	60 – 72	67	
PO2	43 – 66	53	
Na^+	126 – 136	128	
K^+	2,6 – 3,2	2,9	
Cl^-	—	—	
Ca^{++}	1,38 – 1,58	1,41	
Glu	38 – 54	41	
Lac	0,8 – 1,2	0,9	
tHb	—		
O2Hb	—		
COHb	—		
MetHb	—		
HHb	—		

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CVP Level 2 Lot Number 2857

Analyte	CVP Range	CVP Result	Corrective Actions Applied
pH	7.61 - 7,67	7,65	
PCO2	28 - 34	31	
PO2	132 - 154	144	
Na^+	157 - 167	156	
K^+	6,1 - 6,7	6,4	
Cl^-	—	—	
Ca^{++}	0,40 - 0,54	0,43	
Glu	272 - 316	284	
Lac	5,0 - 6,3	5,2	
tHb	—	—	
O2Hb	—	—	
COHb	—	—	
MetHb	—	—	
HHb	—	—	

CVP Level 3 lot Number 3857

Analyte	CVP Range	CVP Result	Corrective Actions Applied
Hct	22 - 26	24	

CVP Level 4 lot Number 4857

Analyte	CVP Range	CVP Result	Corrective Actions Applied
Hct	40 - 46	44	

CVP Level 5 lot Number ——

Analyte	CVP Range	CVP Result	Corrective Actions Applied
tBili			

The analytes that will be analyzed will depend on the instrument, the cartridge type and the instrument configuration.

Notes: _____

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Training

Verify that facility staff has received necessary instruction/training for operation and routine user maintenance.

Operator Name (print)	Title / Position	Signature
Манушкова М.	зав. бакт.	
Хлобе Б. Я.	зав. гр. лабор.	
Журавлёва М.	гр. врач	

Notes: _____

6.0 Operational Supplies

The use of non-IL brand reagents or supplies for testing may cause a clinically significant degradation of performance and results. IL does not assume any obligation or warranty engagement concerning precision and/or accuracy of the measurements nor for any damage to the instrument directly or indirectly resulting from the use of reagents, consumables and expendable supplies different from those produced by IL.

The facility acknowledges that instrument performance is only validated using reagents, consumables and expendable supplies produced by IL.

Facility	Name (print)	Title/Position	Signature
Хлобе	Бусыгина	зав. гр. лабор	

7.0 OQ Acceptance

The Operational Qualification (OQ) has been successfully completed by Instrumentation Laboratory Company or authorized dealer by;

Company	Name (print)	Title / Position	Signature	Date
Labtech Diagnostics	Ilya Nagovskiy	Engineer		10.10.2026

and the facility indicated below accepts installation of the equipment.

Facility	Name	Title / Position	Signature	Date
Хлобе	Бусыгина	зав. гр. лабор.		10.10.202

Notes: _____

The Performance qualification (PQ) is an ongoing process that is part of the user's protocol.